ISLAMIC UNIVERSITY OF TECHNOLOGY

Course Outline and Course Plan

Course Teacher	Dr. Hasan Mahmud	**Designation**	Professor
Department	CSE	**Program**	B.Sc. Engg. SWE
Course Code	CSE 4553	**Course Title**	Machine Learning
Academic Year	2025–2026	**Semester**	Winter
Course Type	Theory	**Requirement**	Elective
Contact Hours	3.00	**Credit Hours**	3.00
Recommended Texts	1. Ethem Alpaydin, *Introduction to Machine Learning*, 3rd ed. , 2014 2. Bishop, C. M., *Pattern Recognition and Machine Learning*, 1st ed. Springer, 2006 3. Ng, A., *Machine Learning Yearning*, Online Version / Draft Edition ed. DeepLearning.AI, 2018		
Prerequisites	Math 4341: Linear Algebra, Math 4441: Probability and Statistics		
Course Homepage	Google Classroom Code: `o2sfduqd`		
Teaching Methods/ Approaches	☑ Lecture ☑ Demonstration ☐ Project	☐ Group Discussion ☑ Problem Solving ☐ Other:	
Teaching Aids	☑ Multimedia ☐ OHP	☑ Board and Marker ☐ Other:	

Course Details

Course Contents (As approved by the Academic Council)

Introduction: Defining machine learning, Scalability, Privacy issues and social impact, Applications in AI, Computer vision, Computer games, Search engines, Marketing, Bioinformatics, Robotics, HCI and Graphics.

Graphical models: Introduction to discrete probability, Inference in Bayesian networks, Maximum likelihood and Bayesian learning Model selection.

Supervised learning: Introduction to continuous probability, Linear regression and classification (least squares and ridge), Model assessment and cross-validation, Introduction to optimization, Nonlinear regression (neural nets and Gaussian processes), and Boosting and feature selection.

Unsupervised learning: Nearest neighbours and K-means, Spectral kernel methods for clustering and semi-supervised learning. The EM algorithm, Mixture models for discrete and continuous data, Temporal methods: hidden Markov models & Kalman filters, Boltzmann machines and random fields, Examples: web mining, collaborative filtering, music and image clustering, automatic, translation, spam filtering, computer games and object recognition.

Neural Network: Fundamentals of Neural Networks, Back-propagation and related training algorithms, Hebbian learning, Cohen-Grossberg learning, The BAM and the Hopfield Memory, Simulated Annealing, Different type of Neural Networks: Counterpropagation, Probabilistic, Radial Basis Function, Generalized Regression, etc, Adaptive Resonance Theory, Dynamic Systems and Neural Control, The Boltzmann

Machine, Self-organizing maps, Spatiotemporal Pattern Classification, The Neocognition, Practical aspects of Neural Networks.

Other forms of learning: Semi-supervised learning, Active learning, Reinforcement learning, Self-taught learning, Evolutionary learning: Genetic algorithm, Genetic programming, CGA.

Course Objectives

- Able to apply fundamental concepts of machine learning techniques in different real-life problem scenarios
- Analyse strengths and weaknesses of different machine learning algorithms and able to devise solutions using different machine learning techniques including supervised, semi-supervised, unsupervised, and reinforcement learnings.
- Design solutions/algorithms for a range of complex engineering problems for real-world machine-learning problems
- Develop or Implement machine learning solutions/algorithms using appropriate tools and libraries

Course Outcomes (COs)

COs	CO Statements	Corresponding POs	Learning Domain and Taxonomy Levels
CO1	Apply fundamental concepts of machine learning in different problem scenarios	PO1	C3
CO2	Analyse sthe trengths and weaknesses of many machine learning algorithms	PO2	C3
CO3	Design solutions/algorithms for a range of real-world machine-learning problems	PO3	C1, C4
CO4	Develop or implement machine learning solutions/ algorithms using appropriate tools and libraries	PO5	C3

Mapping Table of Knowledge Profile, Complex Engineering Problem Solving, and Complex Engineering Activities

K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
✓	✓	✓	✓	✓	✓		✓	✓	✓	✓					✓				

Lecture Plan

Weekly Plan for Course Content	
Week(s)	**Topics**
1	**ML Course Introduction:** Topic overview, assessment, and grading policies. Introduction to Machine Learning, types of learning, steps in developing machine learning applications, overview of different machine learning applications.
2	**Regression analysis:** Linear regression with single and multiple variables, cost function, logistic regression/non-linear regression, regularization
3	**Convexity and non-convexity:** Gradient Descent, Stochastic Gradient Descent, Batch Gradient Descent
4	**Decision tree:** Construction of decision tree, information gain, entropy, splitting dataset.
5	**Probabilistic Classifier:** Discriminative and Generative probabilistic models, Probabilistic classifier: Introduction to probabilistic classifier, Naïve Bayes classifier, Applications of probabilistic classifier

Weekly Plan for Course Content	
Week(s)	**Topics**
6	**Basic Practices in Machine Learning:** Feature engineering, model evaluation techniques, overfitting/underfitting, machine learning model selection and design considerations, hyper-parameters, and performance evaluation.
7	**Classifier performance evaluation:** Confusion matrix, Sensitivity, Specificity, Precision, ROC, AUC, F1-Score and Matthew's correlation coefficient.
8	**K-nearest neighbor:** Intuitive understanding of KNN: How does KNN work? Distance measurement techniques and how to choose the k factor.
9	**Ensembles:** Simple ensembles, max voting, averaging, weighted averaging, Bagging: Random Forest, Ensembles: Boosting technique and algorithms, Adaboost.
10	**Support vector machine (SVM):** What is SVM? How does SVM works? Mathematical formulation of SVM;Kernel Trick; Multi-class classification
11	**Clustering:** Unsupervised learning basics, clustering concepts, K-means/k-medoid clustering, Hierarchical clustering technique: Agglomerative clustering; Density Based Clustering
12	**Dimensionality reduction:** Principal Component Analysis (PCA), Expectation Maximization
13	**Introduction to neural network:** Neural network basics, types of activation functions, feed-forward neural network, and backpropagation neural network.
14	**Introduction to deep neural network:** Deep neural network basics, limitations of neural network, convolution neural network, recurrent neural network, long-short term memory.
15	**Time-series classifier:** Basics of time-series matching, understanding Dynamic Time Warping algorithm. Applications of time-series classifier.

Assessment Strategy

- **Class participation:** Class participation will be assessed with the attendance and impromptu quizzes during class periods
- **Formative assessment:** Time to time formative assessment will be carried out through written quizzes and/or homework assignments
- **Summative assessment:** Two summative assessments will be carried out through the mid and semester final examinations

Distribution of Marks

Module	Percentage (%)
Attendance	10
Quizzes/Assignments/Projects (Continuous Assessment)	30
Mid Semester Examination	30
Semester Final Examination	30
Total	**100**

Instructor Contact Details

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